PHM Derivations

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1 Euler Lagrange

$$q_i = q_i(t)$$
$$L = L(q_i, \dot{q}_i, t).$$

Hamilton's principle

$$S[q] = \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t)\,\mathrm{d}t.$$

Equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0.}$$

So including work

$$\Delta W = \sum_i Q_i\,\Delta q_i.$$

The Lagrange–d'Alembert equation is therefore

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i.}$$

Rayleigh Dissipation w/ function R,

$$Q_i^{\text{diss}} = -\frac{\partial\ R}{\partial \dot{q}_i},$$

so the general Euler-Lagrange equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i^{\text{prop}} + Q_i^{\text{impact}} - \frac{\partial\ R}{\partial \dot{q}_i}.}$$

where

$$Qi = Force_{generalized}$$

2 Proper Time, Lorentz Factor, and Four-Momentum

Spacetime/Minowski interval

$$c^2\,\mathrm{d}\tau^2 = c^2\,\mathrm{d}t^2 - \mathrm{d}x^2.$$

Since $v = \mathrm{d}x/\,\mathrm{d}t$, $v\,\mathrm{d}t = \mathrm{d}x$, substitute in dx

$$c^2\,\mathrm{d}\tau^2 = c^2\,\mathrm{d}t^2 - v^2\,\mathrm{d}t^2 \tag{1}$$

$$= c^2\left(1 - \frac{v^2}{c^2}\right)\,\mathrm{d}t^2. \tag{2}$$

Therefore

$$\boxed{\mathrm{d}\tau = \sqrt{1 - \frac{v^2}{c^2}}\,\mathrm{d}t, \qquad \gamma = \frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{\sqrt{1 - v^2/c^2}}.}$$

Sub $\beta = v/c$,

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}.$$

The four-position and four-velocity are

$$x^\mu = (ct, x, y, z), \qquad U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \gamma(c, \boldsymbol{v}_x, v_y, v_z).$$

The four-momentum is

$$P^\mu = mU^\mu = \left(\frac{E}{c}, \boldsymbol{p}\right),$$

so

$$\boxed{\boldsymbol{p} = \gamma m\boldsymbol{v}, \qquad E = \gamma mc^2.}$$

The constant norm (so it depends on rest mass)

$$P^\mu P_\mu = m^2c^2,$$

or

$$\boxed{E^2 = p^2c^2 + m^2c^4.}$$

Rapidity η = addable relativity velocity, relationship between rapidity and relativstic velocity

$$\beta = \tanh\eta.$$

Hence

$$\boxed{\gamma = \cosh\eta, \qquad \gamma\beta = \sinh\eta, \qquad v = c\tanh\eta.}$$

3 Relativistic (Classical Mechanics see 7.7)

$$S = -mc^2\int \mathrm{d}\tau.$$

Using $\mathrm{d}\tau = \sqrt{1 - v^2/c^2}\ \mathrm{d}t$,

$$S = \int L_{\mathrm{rel}}\,\mathrm{d}t = -mc^2 \int \sqrt{1 - \frac{v^2}{c^2}}\ \mathrm{d}t,$$

so

$$\boxed{L_{\mathrm{rel}} = -mc^2\sqrt{1 - \frac{v^2}{c^2}}.}$$

Macclaurin Approximation

$$\sqrt{1 - z} = 1 - \frac{z}{2} - \frac{z^2}{8} + O(z^3)$$

with $z = v^2/c^2$:

$$L_{\mathrm{rel}} = -mc^2\left(1 - \frac{v^2}{2c^2} - \frac{v^4}{8c^4} + \cdots\right) \tag{3}$$

$$= -mc^2 + \frac{1}{2}mv^2 + \frac{1}{8}m\frac{v^4}{c^2} + \cdots. \tag{4}$$

The canonical momentum is

$$p = \frac{\partial L_{\mathrm{rel}}}{\partial v} \tag{5}$$

$$= -mc^2\left[\frac{1}{2}\left(1 - \frac{v^2}{c^2}\right)^{-1/2}\left(-\frac{2v}{c^2}\right)\right] \tag{6}$$

$$= \frac{mv}{\sqrt{1 - v^2/c^2}}. \tag{7}$$

Therefore

$$\boxed{p = \gamma m v.}$$

The energy function is

$$H = pv - L.$$

Substitution

$$H = \gamma m v^2 + mc^2\sqrt{1 - \frac{v^2}{c^2}} \tag{8}$$

$$= \gamma m v^2 + \frac{mc^2}{\gamma} \tag{9}$$

$$= mc^2\left(\gamma\beta^2 + \frac{1}{\gamma}\right). \tag{10}$$

Since

$$\gamma\beta^2 + \frac{1}{\gamma} = \frac{\beta^2 + 1 - \beta^2}{\sqrt{1 - \beta^2}} = \gamma,$$

we obtain

$$\boxed{H = E = \gamma mc^2.}$$

Equivalently, using $p = \gamma m v$,

$$\boxed{H = \sqrt{p^2c^2 + m^2c^4}.}$$

4 Acceleration and Decceleration

Proper/Coordinate (outside reference frame) acceleration

$$\boxed{\alpha = \gamma^3 \frac{\mathrm{d}v}{\mathrm{d}t}.}$$

Because

$$\frac{\mathrm{d}}{\mathrm{d}t}(\gamma v) = \gamma^3 \frac{\mathrm{d}v}{\mathrm{d}t},$$

constant proper acceleration gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(\gamma v) = \alpha.$$

Starting from rest at $t = 0$,

$$\gamma v = \alpha t.$$

Since $\gamma v = v/\sqrt{1 - v^2/c^2}$,

$$\frac{v^2}{1 - v^2/c^2} = \alpha^2 t^2, \tag{11}$$

$$v^2 \left(1 + \frac{\alpha^2 t^2}{c^2}\right) = \alpha^2 t^2. \tag{12}$$

Thus

$$\boxed{v(t) = \frac{\alpha t}{\sqrt{1 + (\alpha t/c)^2}}.}$$

Integrating,

$$x(t) = \int_0^t \frac{\alpha t'}{\sqrt{1 + (\alpha t'/c)^2}}\,\mathrm{d}t' \tag{13}$$

$$= \frac{c^2}{\alpha}\left[\sqrt{1 + \left(\frac{\alpha t}{c}\right)^2} - 1\right]. \tag{14}$$

Therefore

$$\boxed{x(t) = \frac{c^2}{\alpha}\left[\sqrt{1 + (\alpha t/c)^2} - 1\right].}$$

In rapidity,

$$\alpha = c\frac{\mathrm{d}\eta}{\mathrm{d}\tau}.$$

For constant α and initial rest,

$$\eta = \frac{\alpha \tau}{c}.$$

Using $\mathrm{d}t/\,\mathrm{d}\tau = \cosh\eta$ and $\mathrm{d}x/\,\mathrm{d}\tau = c\sinh\eta$,

$$t(\tau) = \int_0^\tau \cosh\left(\frac{\alpha\tau'}{c}\right)\,\mathrm{d}\tau' = \frac{c}{\alpha}\sinh\left(\frac{\alpha\tau}{c}\right), \tag{15}$$

$$x(\tau) = \int_0^\tau c \sinh\left(\frac{\alpha\tau'}{c}\right) \mathrm{d}\tau' = \frac{c^2}{\alpha}\left[\cosh\left(\frac{\alpha\tau}{c}\right) - 1\right]. \tag{16}$$

$$\boxed{\begin{aligned} t(\tau) &= \frac{c}{\alpha}\sinh\left(\frac{\alpha\tau}{c}\right), \\ x(\tau) &= \frac{c^2}{\alpha}\left[\cosh\left(\frac{\alpha\tau}{c}\right) - 1\right], \\ v(\tau) &= c\tanh\left(\frac{\alpha\tau}{c}\right). \end{aligned}}$$

Accelerate for D/2, brake for D/2

$$\frac{D}{2} = \frac{c^2}{\alpha}(\gamma_h - 1),$$

so

$$\boxed{\gamma_h = 1 + \frac{\alpha D}{2c^2}, \qquad \eta_h = \operatorname{arcosh}\gamma_h.}$$

For a separate constant force, constant mass

$$\frac{\mathrm{d}}{\mathrm{d}t}(\gamma m_r v) = F_0.$$

Starting from rest,

$$\gamma m_r v = F_0 t.$$

Integrate

$$\boxed{x(t) = \frac{m_r c^2}{F_0}\left[\sqrt{1 + (F_0 t/m_r c)^2} - 1\right].}$$

5 Changing mass

For an instantaneous spacecraft mass $m(t)$,

$$p = \gamma m v.$$

Differentiate all three

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \dot\gamma m v + \gamma \dot m v + \gamma m \dot v. \tag{17}$$

Since

$$\dot\gamma = \frac{\mathrm{d}}{\mathrm{d}t}\left(1 - \frac{v^2}{c^2}\right)^{-1/2} = \gamma^3 \frac{v\dot v}{c^2},$$

we have

$$\frac{\mathrm{d}p}{\mathrm{d}t} = m\gamma^3 \frac{v^2 \dot v}{c^2} + \gamma v \dot m + \gamma m \dot v \tag{18}$$

$$= m\gamma\dot v\left(1 + \gamma^2 \frac{v^2}{c^2}\right) + \gamma v \dot m. \tag{19}$$

Using $1 + \gamma^2\beta^2 = \gamma^2$,

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}t} = m\gamma^3\dot{v} + \gamma v\dot{m}.}$$

A raw open-system force equation can be written

$$m\gamma^3\dot{v} + \gamma v\dot{m} = Q_{\text{prop}} + F_{\text{ext}}.$$

$$\boxed{m\gamma^3\dot{v} = T + F_{\text{ext}}.}$$

Photon beam with Power P carries momentum at a rate P/c. With collimation efficiency e,

$$T = e\frac{P_\gamma}{c}.$$

If a fraction ϵ of consumed rest energy becomes useful energy,

$$P = -e\epsilon c^2\frac{\mathrm{d}m}{\mathrm{d}\tau}.$$

$$\boxed{T = -e\epsilon c\frac{\mathrm{d}m}{\mathrm{d}\tau}.}$$

For constant proper acceleration, $T = m\alpha$, so

$$m\alpha = -\chi\epsilon c\frac{\mathrm{d}m}{\mathrm{d}\tau}.$$

Separate variables:

$$\frac{\mathrm{d}m}{m} = -\frac{\alpha}{\chi\epsilon c}\,\mathrm{d}\tau.$$

Integrate from (m_i, τ_i) to (m_f, τ_f),

$$\ln\frac{m_f}{m_i} = -\frac{\alpha\Delta\tau}{\chi\epsilon c}.$$

$\Delta\eta = \alpha\Delta\tau/c$,

$$\boxed{\frac{m_i}{m_f} = \exp\left(\frac{\Delta\eta}{\chi\epsilon}\right).}$$

For a acceleration and braking trip,

$$\boxed{R_{\text{total}} = R_h^2 = \exp\left(\frac{2\eta_h}{\chi\epsilon}\right).}$$

6 Gravity and Radiation

$$r_s(x) = r_{s0} + x, \qquad r_T(x) = D + r_{T0} - x.$$

The gravitational potential energy is

$$U_g(x) = -\frac{GM_s m}{r_s(x)} - \frac{GM_T m}{r_T(x)}.$$

Lagrangian

$$L_g = -U_g = \frac{GM_s m}{r_s} + \frac{GM_T m}{r_T}.$$

The force $F_g = \partial L_g / \partial x$:

$$\frac{\partial}{\partial x}\left(\frac{GM_s m}{r_s}\right) = -\frac{GM_s m}{r_s^2}, \tag{20}$$

$$\frac{\partial}{\partial x}\left(\frac{GM_T m}{r_T}\right) = +\frac{GM_T m}{r_T^2}. \tag{21}$$

Hence

$$\boxed{F_g = -\frac{GM_s m}{r_s^2} + \frac{GM_T m}{r_T^2}.}$$

For a star of luminosity L, area A, and radiation coefficient C_R,

$$F_{\text{rad}} = C_R \frac{AL}{4\pi c r^2}.$$

Define

$$K_* \equiv \frac{C_R A L}{4\pi c}.$$

An effective potential $U_{\text{rad}} = K_*/r$ gives a force

$$\boxed{F_{\text{rad}} = \frac{K_s}{r_s^2} - \frac{K_T}{r_T^2}.}$$

The corresponding Lagrangian terms are

$$L_{\text{rad}} = -\frac{K_s}{r_s} - \frac{K_T}{r_T}.$$

7 Relativistic Gas and Dust Drag

The number of collisions $= \gamma n$, the collision rate through front area $A = \gamma n A v$, and the incident particle momentum $= \gamma m_p v$. Particle per area traveled $=$

$$F \sim -(\gamma n A v)(\gamma m_p v).$$

Writing $\rho = n m_p$ and elasticity/momentum transfer $= C$,

$$\boxed{F_{\text{drag}} = -C\rho A \gamma^2 v|v|.}$$

For gas and smooth dust together,

$$\boxed{F_{g+d} = -\left(C_g \rho_g + C_d \rho_d\right) A \gamma^2 v|v|.}$$

The power from drag

$$\boxed{P_{\text{drag}} = |F_{\text{drag}} v|.}$$

drag force from Rayleigh Dissapation

$$-\frac{\partial R_{\text{drag}}}{\partial v} = -C\rho A\gamma^2 v|v|.$$

For $v > 0$,

$$R_{\text{drag}} = C\rho A \int_0^v \frac{u^2}{1 - u^2/c^2}\,\mathrm{d}u. \tag{22}$$

Set $\beta_u = u/c$, so $u = c\beta_u$ and $\mathrm{d}u = c\,\mathrm{d}\beta_u$:

$$R_{\text{drag}} = C\rho A c^3 \int_0^\beta \frac{\beta_u^2}{1 - \beta_u^2}\,\mathrm{d}\beta_u. \tag{23}$$

Use

$$\frac{\beta_u^2}{1 - \beta_u^2} = -1 + \frac{1}{1 - \beta_u^2}.$$

Then

$$\int_0^\beta \frac{\beta_u^2}{1 - \beta_u^2}\,\mathrm{d}\beta_u = [-\beta_u + \operatorname{artanh}\beta_u]_0^\beta \tag{24}$$

$$= \operatorname{artanh}\beta - \beta. \tag{25}$$

Both signs of v,

$$\boxed{R_{\text{drag}} = C\rho A c^3 \left(\operatorname{artanh}|\beta| - |\beta|\right).}$$

Double Checking

$$\frac{\partial R_{\text{drag}}}{\partial v} = C\rho A c^3 \frac{\operatorname{sgn}(v)}{c}\left(\frac{1}{1 - \beta^2} - 1\right) \tag{26}$$

$$= C\rho A c^2 \operatorname{sgn}(v) \frac{\beta^2}{1 - \beta^2} \tag{27}$$

$$= C\rho A\gamma^2 v|v|. \tag{28}$$

Therefore

$$-\frac{\partial R_{\text{drag}}}{\partial v} = -C\rho A\gamma^2 v|v|.$$

For $|\beta| \ll 1$,

$$\operatorname{artanh}|\beta| - |\beta| = \frac{|\beta|^3}{3} + O(|\beta|^5),$$

so

$$\boxed{R_{\text{drag}} \longrightarrow \frac{C\rho A}{3}|v|^3,}$$

whose derivative = ordinary quadratic drag.

8 Linear Damping and Friction (See Textbook)

For linear damping $F_d = -c\dot{q}$

$$\boxed{R_{\text{visc}} = \frac{1}{2}c\dot{q}^2,}$$

because

$$-\frac{\partial R_{\text{visc}}}{\partial \dot{q}} = -c\dot{q}.$$

For Coulomb friction $F_C = -F_0 \operatorname{sgn}(\dot{q})$

$$\boxed{R_C = F_0|\dot{q}|,}$$

away from $\dot{q} = 0$,

$$-\frac{\partial R_C}{\partial \dot{q}} = -F_0 \operatorname{sgn}(\dot{q}).$$

For a damped oscillator,

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2, \qquad R = \frac{1}{2}c\dot{x}^2.$$

The forced Euler–Lagrange equation gives

$$\boxed{m\ddot{x} + c\dot{x} + kx = 0.}$$

dampened natural frequency/ zeta is dampening ratio

$$\boxed{\omega_n = \sqrt{\frac{k}{m}}, \qquad \zeta = \frac{c}{2m\omega_n} = \frac{c}{2\sqrt{km}}.}$$

Thus

$$\boxed{c = 2\zeta\sqrt{km}.}$$

A discrete impact at time t_k is represented by

$$Q_{\text{impact}}(t) = J_k\delta(t - t_k).$$

Integrating the momentum equation across +- (small time t_k,

$$\int_{t_k^-}^{t_k^+} \dot{p}\,\mathrm{d}t = \int_{t_k^-}^{t_k^+} J_k\delta(t - t_k)\,\mathrm{d}t, \tag{29}$$

$$p(t_k^+) - p(t_k^-) = J_k. \tag{30}$$

Therefore

$$\boxed{\Delta p = J_k.}$$

In vector form,

$$\boxed{\boldsymbol{Q}_{\text{impact}}(t) = \sum_k \boldsymbol{J}_k\delta(t - t_k),}$$

and an impact at body-fixed lever arm $\boldsymbol{r}_k$ produces

$$\boxed{\boldsymbol{\tau}_{\text{impact}}(t) = \sum_k \boldsymbol{r}_k \times \boldsymbol{J}_k\,\delta(t - t_k).}$$

9 Tether and Centrifuge

For N_j parallel cables of diameter d_c, Young's modulus E, and undeformed length r_j,

$$A_c = \frac{\pi d_c^2}{4}, \qquad \boxed{k_j = \frac{N_j E A_c}{r_j}.}$$

x_j radial extension of spinning mass (x)M_j, and let ϑ be the centrifuge angle. The radius is

$$R_j = r_j + x_j.$$

$$v_j^2 = \dot{(x_j)}^2 + R_j^2 \dot{\vartheta}^2.$$

Thus

$$T_j = \frac{1}{2} M_j \left[\dot{(x_j)}^2 + (r_j + x_j)^2 \dot{\vartheta}^2 \right],$$

while

$$U_j = \frac{1}{2} k_j x_j^2.$$

The centrifuge Lagrangian is

$$\boxed{L_{\text{cent}} = \sum_{j=h,c} \left[\frac{1}{2} M_j \dot{(x_j)}^2 + \frac{1}{2} M_j (r_j + x_j)^2 \dot{\vartheta}^2 - \frac{1}{2} k_j x_j^2 \right].}$$

For x_j,

$$\frac{\partial L}{\partial \dot{(x_j)}} = M_j \dot{(x_j)}, \qquad \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{(x_j)}} = M_j \ddot{(x_j)},$$

and

$$\frac{\partial L}{\partial (x_j)} = M_j (r_j + (x_j)) \dot{\vartheta}^2 - k_j (x_j).$$

Hence

$$M_j \ddot{(x_j)} - M_j (r_j + (x_j)) \dot{\vartheta}^2 + k_j (x_j) = Q_{(x_j)}.$$

With viscous damping, Coulomb friction, and impact forces,

$$Q_{(x_j)} = -c_j \dot{(x_j)} - F_{C,j} \operatorname{sgn}(\dot{(x_j)}) + Q_{(x_j)}^{\text{impact}}.$$

added together

$$\boxed{M_j \ddot{(x_j)} + c_j \dot{(x_j)} + k_j (x_j) + F_{C,j} \operatorname{sgn}(\dot{(x_j)}) = M_j (r_j + (x_j)) \dot{\vartheta}^2 + Q_{(x_j)}^{\text{impact}}.}$$

At rotating equilibrium,

$$\dot{(x_j)} = \ddot{(x_j)} = 0, \qquad \dot{\vartheta} = \omega,$$

so

$$k_j \xi_{j,\text{eq}} = M_j (r_j + \xi_{j,\text{eq}}) \omega^2.$$

Collecting the extension terms,

$$(k_j - M_j \omega^2) \xi_{j,\text{eq}} = M_j r_j \omega^2.$$

Thus

$$\boxed{\xi_{j,\mathrm{eq}} = \frac{M_j r_j \omega^2}{k_j - M_j \omega^2}.}$$

The centrifuge moment of inertia is

$$\boxed{I_{\mathrm{cent}}(\xi_h, \xi_c) = \sum_{j=h,c} M_j (r_j + (x_j))^2.}$$

Since

$$\frac{\partial L}{\partial \dot{\vartheta}} = I_{\mathrm{cent}} \dot{\vartheta},$$

we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(I_{\mathrm{cent}} \dot{\vartheta}\right) = I_{\mathrm{cent}} \ddot{\vartheta} + \dot{I}_{\mathrm{cent}} \dot{\vartheta}.$$

Therefore

$$\boxed{I_{\mathrm{cent}} \ddot{\vartheta} + \dot{I}_{\mathrm{cent}} \dot{\vartheta} = \tau_{\mathrm{motor}} - b_\vartheta \dot{\vartheta} - \tau_C \operatorname{sgn}(\dot{\vartheta}) + \tau_{\mathrm{impact}}.}$$

10 Rigid-Body Rotation and Attitude

Let $\boldsymbol{\omega}$ be the body angular velocity and $\boldsymbol{I}$ the inertia . The Lagrangian is

$$\boxed{L_{\mathrm{att}} = \frac{1}{2} \boldsymbol{\omega}^T \boldsymbol{I} \boldsymbol{\omega} - U_{gg} - U_{\mathrm{other}}.}$$

Gravity Gradiant (gravity difference between ends of ship)

$$\boxed{U_{gg} = -\frac{\mu}{2r^3} \left[\mathrm{tr}(\boldsymbol{I}) - 3 \widehat{\boldsymbol{r}}^T \boldsymbol{I} \widehat{\boldsymbol{r}}\right].}$$

Its associated torque is

$$\boxed{\boldsymbol{\tau}_{gg} = \frac{3\mu}{r^3} \widehat{\boldsymbol{r}} \times (\boldsymbol{I} \widehat{\boldsymbol{r}}).}$$

The body-frame angular-momentum balance is

$$\boxed{\boldsymbol{I} \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\boldsymbol{I} \boldsymbol{\omega}) = \boldsymbol{\tau}_{\mathrm{engine}} + \boldsymbol{\tau}_{\mathrm{rad}} + \boldsymbol{\tau}_{gg} + \boldsymbol{\tau}_{\mathrm{EM}} + \boldsymbol{\tau}_{\mathrm{impact}} - \boldsymbol{B}_\omega \boldsymbol{\omega}.}$$

11 Electromagnetic Coupling from ship

For the charge-coupling terms, in Cartesian component form,

$$\frac{\partial L_{\mathrm{EM}}}{\partial \dot{r}_i} = Q A_i,$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L_{\mathrm{EM}}}{\partial \dot{r}_i} = Q \frac{\partial A_i}{\partial t} + Q \dot{r}_j \frac{\partial A_i}{\partial r_j}.$$

Also,

$$\frac{\partial L_{\mathrm{EM}}}{\partial r_i} = Q\dot{r}_j \frac{\partial A_j}{\partial r_i} - Q\frac{\partial \phi}{\partial r_i} + \frac{\partial}{\partial r_i}(\boldsymbol{\mu}_b \cdot \boldsymbol{B}).$$

The Euler–Lagrange equation therefore gives

$$F_i = Q\left[-\frac{\partial A_i}{\partial t} - \frac{\partial \phi}{\partial r_i} + \dot{r}_j\left(\frac{\partial A_j}{\partial r_i} - \frac{\partial A_i}{\partial r_j}\right)\right] + \frac{\partial}{\partial r_i}(\boldsymbol{\mu}_b \cdot \boldsymbol{B}). \tag{31}$$

Using

$$\boldsymbol{E} = -\nabla\phi - \frac{\partial \boldsymbol{A}}{\partial t}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A},$$

we obtain

$$\boxed{\boldsymbol{F}_{\mathrm{EM}} = Q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) + \nabla(\boldsymbol{\mu}_b \cdot \boldsymbol{B}),}$$

with magnetic torque

$$\boxed{\boldsymbol{\tau}_{\mathrm{EM}} = \boldsymbol{\mu}_b \times \boldsymbol{B}.}$$

12 Combined Dissipation Function

Combining smooth gas drag, smooth dust drag, tether damping and friction, centrifuge bearing losses, and attitude damping gives

$$\boxed{\begin{aligned} R = {} & (C_g\rho_g + C_d\rho_d)Ac^3\left(\operatorname{artanh}|\beta| - |\beta|\right) \\ & + \frac{1}{2}c_h\dot{\xi}_h^2 + \frac{1}{2}c_c\dot{\xi}_c^2 + F_{C,h}|\dot{\xi}_h| + F_{C,c}|\dot{\xi}_c| \\ & + \frac{1}{2}b_\vartheta\dot{\vartheta}^2 + \tau_{C,\vartheta}|\dot{\vartheta}| + \frac{1}{2}\boldsymbol{\omega}^T\boldsymbol{B}_\omega\boldsymbol{\omega}. \end{aligned}}$$

Each smooth dissipative generalized force is

$$Q_i^{\mathrm{diss}} = -\frac{\partial \mathcal{R}}{\partial \dot{q}_i}.$$

Discrete impacts remain separate delta-function generalized forces.

13 Final Combined Lagrangian

Let $\boldsymbol{r}$ be the center-of-mass position, with axial coordinate x, and let $m = m(\tau)$ be supplied by the propulsion subsystem. The complete conservative effective Lagrangian is

$$\boxed{\begin{aligned} L = {} & -m(\tau)c^2\sqrt{1 - \frac{\dot{\boldsymbol{r}}^{\,2}}{c^2}} + \frac{GM_s m}{r_s} + \frac{GM_T m}{r_T} - \frac{K_s}{r_s} - \frac{K_T}{r_T} \\ & + Q\boldsymbol{A}\cdot\dot{\boldsymbol{r}} - Q\phi + \boldsymbol{\mu}_b \cdot \boldsymbol{B} + \frac{1}{2}\boldsymbol{\omega}^T\boldsymbol{I}\boldsymbol{\omega} - U_{gg} \\ & + \sum_{j=h,c}\left[\frac{1}{2}M_j\dot{(x_j)}^2 + \frac{1}{2}M_j(r_j + (x_j))^2\dot{\vartheta}^2 - \frac{1}{2}k_j(x_j)^2\right]. \end{aligned}}$$

Here

$$\boxed{\begin{aligned} r_s &= r_{s0} + x, \\ r_T &= D + r_{T0} - x, \\ K_s &= \frac{C_R A L_s}{4\pi c}, \\ K_T &= \frac{C_R A L_T}{4\pi c}. \end{aligned}}$$

The propulsion subsystem supplies $m(\tau),\ \mathrm{d}m/\,\mathrm{d}\tau$, thrust, and exhaust power; it is not inserted as an additional conservative potential.

14 Final Equations of Motion

The general forced Euler–Lagrange equation is

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i^{\mathrm{prop}} + Q_i^{\mathrm{impact}} - \frac{\partial R}{\partial \dot{q}_i}.}$$

For axial translation,

$$\boxed{\begin{aligned} m\gamma^3 \ddot{x} = T_x &- \frac{GM_s m}{r_s^2} + \frac{GM_T m}{r_T^2} + \frac{K_s}{r_s^2} - \frac{K_T}{r_T^2} \\ &+ Q[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}]_x - (C_g \rho_g + C_d \rho_d) A \gamma^2 v |v| \\ &+ \sum_k J_{k,x} \delta(t - t_k). \end{aligned}}$$

A convenient proper-time rapidity form is

$$\boxed{\frac{\mathrm{d}t}{\mathrm{d}\tau} = \cosh\eta, \qquad \frac{\mathrm{d}x}{\mathrm{d}\tau} = c\sinh\eta, \qquad mc\frac{\mathrm{d}\eta}{\mathrm{d}\tau} = T_x + F_{\mathrm{external}}, \qquad \frac{\mathrm{d}m}{\mathrm{d}\tau} = -\frac{T}{\chi \epsilon c}.}$$

The internal radial equations are

$$\boxed{M_j \ddot{(x_j)} + c_j \dot{(x_j)} + k_j(x_j) + F_{C,j}\,\mathrm{sgn}(\dot{(x_j)}) = M_j(r_j + (x_j))\dot{\vartheta}^2 + Q_{(x_j)}^{\mathrm{impact}}.}$$

The centrifuge spin equation is

$$\boxed{I_{\mathrm{cent}}\ddot{\vartheta} + \dot{I}_{\mathrm{cent}}\dot{\vartheta} = \tau_{\mathrm{motor}} - b_\vartheta \dot{\vartheta} - \tau_C\,\mathrm{sgn}(\dot{\vartheta}) + \tau_{\mathrm{impact}}.}$$

The rigid-body attitude equation is

$$\boxed{\boldsymbol{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\boldsymbol{I}\boldsymbol{\omega}) = \boldsymbol{\tau}_{\mathrm{engine}} + \boldsymbol{\tau}_{\mathrm{rad}} + \boldsymbol{\tau}_{gg} + \boldsymbol{\tau}_{\mathrm{EM}} + \boldsymbol{\tau}_{\mathrm{impact}} - \boldsymbol{B}_\omega \boldsymbol{\omega}.}$$